

Endomorphism Rings of Elliptic Curves

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1 Preliminary

1.1 Invariants on elliptic curves

1.1.1 j -invariant

1.2 Endomorphism ring

Notice that $[n] : E \rightarrow E \quad P \mapsto [n](P)$ is an endomorphism of E . By varying n , there is a family of endomorphisms, which induces a ring homomorphism

$$\mathbb{Z} \rightarrow \text{End}(E) \quad n \mapsto [n]$$

This is also natural in category, because $\mathbb{Z}$ is the initial object for the category of commutative ring.

1.2.1 Structures and properties of $\text{End}(E)$

Corollary 1.1: Properties of isogeny

Let $\phi : E_1 \rightarrow E_2$ be a non-zero isogeny.

(1) $\forall Q \in E_2$, the fiber of Q equals the separable degree of ϕ , namely $|\phi^{-1}(Q)| = \deg_s \phi$.

$\forall P \in E_1$, the $e_\phi(P)$ equals the inseparable degree of ϕ , namely $e_\phi(P) = \deg_i \phi$.

(2) Let $\tau_P : E \rightarrow E \quad B \mapsto B + P$ be a translation by P map. Then, the map

$$\ker \phi \rightarrow \text{Aut}(\overline{K}(E_1)/\phi_*\overline{K}(E_2)) \quad P \mapsto \tau_P^*$$

is an isomorphism.

(3) If ϕ is separable, then ϕ is unramified and $|\ker \phi| = \deg \phi$. In this case, $\overline{K}(E_1)/\phi_*\overline{K}(E_2)$ is a Galois extension.

1.2.2 Properties of dual isogeny

Corollary 1.2: Properties of dual isogeny

Let $\phi : E_1 \rightarrow E_2$ be a nonconstant isogeny of degree $\deg \phi$. Then,

(1) $\exists!$ isogeny $\hat{\phi} : E_2 \rightarrow E_1$ satisfying $\hat{\phi} \circ \phi = [\deg \phi]$.

Remark Especially, when $E_1 = E_2 = E$, the dual isogeny is like a self-adjoint operation: Every isogeny $\phi \in \text{End}(E)$ has a dual isogeny $\hat{\phi} \in \text{End}(E)$ satisfying

$$\phi \circ \hat{\phi} = \hat{\phi} \circ \phi = [\deg \phi]$$

2 Tame world of characteristic 0

2.1 over $\mathbb{C}$

For an elliptic curve E over $\mathbb{C}$, there is an structure theorem

$$E(\mathbb{C}) \cong \mathbb{C}/\Lambda$$

This comes from the map

$$\mathbb{C} \rightarrow E(\mathbb{C}) \quad z \mapsto [\wp(z) : \wp'(z) : 1]$$

via the Weierstrass $\wp$ -function.

2.2 Complex multiplication

Let E_1 and E_2 be two elliptic curves over $\mathbb{C}$ and $\phi : E_1 \rightarrow E_2$ be a non-zero isogeny. By virtue of Riemann surfaces, this ϕ is a finite unramified regular covering. **need more**

If E_1 is a covering space of E_2 , then there

For any elliptic curve E over $\mathbb{C}$, its universal covering space $\widetilde{E(\mathbb{C})} = \mathbb{C}$ is the whole complex plane. As a consequence, there is a natural covering map $\pi : \mathbb{C} \rightarrow E(\mathbb{C})$.

From the above-mentioned structure theorem, $E(\mathbb{C}) \cong \mathbb{C}/\Lambda$.

In addition, $E(\mathbb{C}) = \widetilde{E(\mathbb{C})}/\pi_1(E, \mathcal{O}_1)$. Comparing them, the first fundamental group of E , $\pi_1(E, \mathcal{O}_1)$, is $\pi_1(E) \cong \Lambda$. This makes sense, because $E(\mathbb{C})$ is a torus and the fundamental group of a torus with respect to any point is $\cong \mathbb{Z}^2$. By **what**, any endomorphism $\phi \in \text{End}(E)$, as a continuous map, can be lifted to a map between their universal coverings. Namely, ϕ can be lifted to a linear map

$$\tilde{\phi} : \mathbb{C} \rightarrow \mathbb{C} \quad z \mapsto \alpha z$$

How can this $\tilde{\phi}$ descend to the base space E ? Not every linear map (every $\alpha \in \mathbb{C}$) $\tilde{\phi}$ can descend to the base space E . To make it well-defined in the quotient space, $\tilde{\phi}$ must keep the structure of Λ , i.e. α should satisfy

$$\alpha\Lambda \subseteq \Lambda$$

If Λ is just some mediocre lattice (for example, some lattices that lose more symmetry), then the only operation that keeps $\alpha\Lambda$ a sublattice of Λ is scaling.

If Λ has some higher order **not well-defined**, for example, a lattice of squares or a lattice of hexagons, then there are more operations than only scaling that can keep $\alpha\Lambda$ be a sublattice of Λ , say rotation operations. The complex numbers occur and correspond to rotation operations. In this case, we say that E has complex multiplication. What property should E with complex multiplication own?

Suppose that $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$, with $\omega_1, \omega_2 \in \mathbb{C}$ being linearly independent in $\mathbb{C}$. To consider $\alpha \cdot \Lambda$, it suffices to consider how α act on the basis ω_1 and ω_2 of Λ . Since $\alpha\Lambda \subseteq \Lambda$, we have

$$\begin{aligned} \alpha \cdot \omega_1 &= a\omega_1 + b\omega_2 \\ \alpha \cdot \omega_2 &= c\omega_1 + d\omega_2 \end{aligned}$$

with $a, b, c, d \in \mathbb{Z}$.

Let $\tau := \frac{\omega_2}{\omega_1}$. Then, we have the equation $\tau(a + b\tau) = c + d\tau$, i.e.

$$b\tau^2 + (a - d)\tau - c = 0 \tag{1}$$

where $b \neq 0$ because we are discussing the case that α is not a scaling (or not trivial). Notice that $\tau \in \mathcal{H}$, τ lies in the upper half plane. So, the equation (1) has no real roots. Therefore, its discriminant $\Delta = (a - d)^2 + 4bc < 0$.

Theorem 2.1: Classification for $\text{End}(E)$ on elliptic curves over $\mathbb{C}$

Let E be an elliptic curve over $\mathbb{C}$. Then, the structures of $\text{End}(E)$ have only 2 possibilities:

- (1) $\text{End}(E) \cong \mathbb{Z}$.
- (2) $\text{End}(E) \cong \mathcal{O}$, where $\mathcal{O}$ is the order of some quadratic field $K = \mathbb{Q}(\sqrt{-D})$ for some $D > 0$. In this case, we say that E has a complex multiplication.

Remark There are infinitely many **classes** of E with complex multiplication. But, there are only two cases for E that their automorphism groups $\text{Aut}(E)$ are strictly larger than $\{\pm 1\}$. These two cases are E with $j(E) = 1728$ and E with $j(E) = 0$.

For $j(E) = 1728$, $\text{End}(E) \cong \mathbb{Z}[i]$. For $j(E) = 0$, $\text{End}(E) = \mathbb{Z}[\omega]$.

2.3 Local field: $\mathbb{R}$

2.4 Local fields: $\mathbb{Q}_p$

2.4.1 Structure of E over $\mathbb{Q}_p$

When E is defined over $\mathbb{C}$, we have a nice structure $E(\mathbb{C}) \cong \mathbb{C}/\Lambda$. We would like to ask that whether there is a similar structure theorem for E over $\mathbb{Q}_p$, like $E(\mathbb{Q}_p) \cong \mathbb{Q}_p/\Lambda$, where Λ is a discrete subgroup of $\mathbb{Q}_p$?

Before delving into deep, we first unveil the structure of additive discrete subgroup of $\mathbb{Q}_p$:

Theorem 2.2: Additive discrete subgroups of $\mathbb{Q}_p$

The only additive discrete subgroup of $\mathbb{Q}_p$ is $\{0\}$.

Proof: Let Γ be an additive discrete subgroup of $\mathbb{Q}_p$. Consider the open set $\mathbb{Z}_p$ of $\mathbb{Q}_p$ at the identity element 0. Then, $\Gamma \cap \mathbb{Z}_p$ is finite because it is a discrete subgroup of the compact group $\mathbb{Z}_p$. $\square$

This theorem alerts that it is not possible to get a non-trivial additive discrete subgroup of $\mathbb{Q}_p$ and use it to represent $E(\mathbb{Q}_p)$.

To continue this idea, we go back to the $E(\mathbb{C})$ case to see that if there is any other solutions? Notice that in $\mathbb{C}$ we merely use the addition of $\mathbb{C}$. Is it possible to use the multiplication of $\mathbb{C}$ to get a structure representative of $E(\mathbb{C})$?

Tate [ref](#) gave a solution as follows.

In $\mathbb{C}$, there is a group isomorphism

$$\varphi : (\mathbb{C}, +) \rightarrow (\mathbb{C}^\times, \cdot) \quad z \mapsto e^{2\pi iz}$$

This isomorphism maps Λ to $\varphi(\Lambda) = \{e^{2\pi i n_1 \omega_1} \cdot e^{2\pi i n_2 \omega_2} : n_1, n_2 \in \mathbb{Z}\}$. Substituting $\tau = \frac{\omega_2}{\omega_1}$, let $q = e^{2\pi i \tau}$. Then, $\Lambda \cong \varphi(\Lambda) = q^\mathbb{Z}$. In this manner, $E(\mathbb{C})$ has a new way to be depicted:

$$E(\mathbb{C}) \cong \mathbb{C}^\times / q^\mathbb{Z}$$

This manner enlightens us(also Tate) how to yield a structure theorem for $E(\mathbb{Q}_p)$. We shall find an element $q \in \mathbb{Q}_p$ with $|q|_p < 1$, to get a group quotienting $\mathbb{Q}_p^\times$ by the cyclic subgroup $q^\mathbb{Z}$, $\mathbb{Q}_p^\times / q^\mathbb{Z}$. This is an elliptic curve, called *Tate curve*.

2.4.2 Tate curves and the endomorphism ring

2.4.3 Finite extensions of $\mathbb{Q}_p$

2.4.4 Algebraic closure of $\mathbb{Q}_p$

2.5 Glocal fields

3 Function fields

4 Finite fields

4.1 First perspective(Hasse)

Suppose E is an elliptic curve over a finite field $\mathbb{F}_p$. On $\overline{\mathbb{F}_p}$, there are much more points than $E(\mathbb{F}_p)$. How can we characterize those points on $E(\overline{\mathbb{F}_p})$?

The coordinate of $P = (x, y)$ must satisfy that $x^p = x$ and $y^p = y$. To get (x^p, y^p) , one uses the map π_p induced by Frobenius map Frob_p such that

$$\pi_p(x, y) = (\text{Frob}_p(x), \text{Frob}_p(y))$$

In this language, the question of searching points on $E(\overline{\mathbb{F}_p})$ lying in $E(\mathbb{F}_q)$ is transformed into the equation $\pi_p(P) = P$. Hence,

$$(\text{id} - \pi_p)(P) = 0 \stackrel{\dagger}{\Leftrightarrow} ([1] - \pi_p)(P) = 0 \Leftrightarrow P \in \ker([1] - \pi_p)$$

$\dagger$: id is exactly the map $[1]$.

$[1] - \pi_p$ is separable [!!!](#). Hence, by [1.1](#),

$$\deg([1] - \pi_p) \stackrel{1.1}{=} |\ker([1] - \pi_p)| = |E(\mathbb{F}_p)|$$

$$\pi_p \circ \hat{\pi}_p = [\deg([1] - \pi_p)]$$

4.2 Second perspective(Deuring): From $\mathbb{Q}_p$ to $\mathbb{F}_p$

4.2.1 Reduction of $E(\mathbb{Q}_p)$ to $E(\mathbb{F}_p)$

4.2.2 Deuring injection map

4.3 Third perspective(Tate, Grothendieck)

4.3.1 Étale topology

In classical complex algebraic geometry, a complex manifold has a Euclidean topology from $\mathbb{C}$.

However, when it comes to the geometry studying algebraic curves over $\mathbb{F}_q$. There was barely Zariski topology

associated to an algebraic variety at the very beginning. Zariski topology is very coarse in the sense that every open set is almost the whole space. Especially for an algebraic curve, a Zariski topology on it is just the whole curve except finitely many points on it. This means that every open set is too 'big' to catch some tiny local topology of that curve. As a consequence, it is not able to construct a fine enough cohomology on it (such as the cohomology theory required by the Weil's conjecture).

To find a finer open sets, Grothendieck turned to the inverse theorem in the differential geometry and complex geometry:

Definition 4.1: Étale morphisms

For a morphisms between two schemes X and Y , a morphism $f : Y \rightarrow X$ is **étale** $\Leftrightarrow$

- (1) f is flat.
- (2) f is unramified.

In the classic topology, an open set U is a subset of X . Open sets can also be viewed as an inclusion map $i : U \hookrightarrow X$. This is a relative point of view. Inspired by such paradigm shift, why not define open sets to be morphisms? This implies the following definition, which has the same role as open sets in classic topology.

Definition 4.2: Étale site (open sets)

Remark In classic topology, the concept of open sets is prior to the concept of continuous map. For étale case, the dependency reverses. Étale sites(open sets) depends on étale morphisms.

4.3.2 Étale cohomology

4.3.3 Tate's Isogeny theorem

4.4 Remedy(Dieudonné, Grothendieck)

4.4.1 Crystalline cohomology

4.4.2 fppf

4.5 Ordinary elliptic curves

4.6 Supersingular elliptic curves

5 Post-quantum cryptography: Isogeny cryptography